

ISLR Chapter 3

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Chapter 3

Simple Linear Regression

- It involves predicting a response Y using a single predictor variable X
- It assumes a linear relation: $Y = \beta_0 + \beta_1 X$
- β_0 and β_1 are called parameters or estimators
- Some notation: The estimated parameters are written as $\hat{\beta}_i$ and the predicted response is written as $\hat{y}$
- We want our model to be close to our data. To ensure reasonable closeness, we try minimizing the 'RSS' (Residual Sum of Squares). The error (e_i) in the i th prediction (for the i th observation) is given by $y_i - \hat{y}_i$.
- $RSS = \sum_{i=1}^n e_i^2$
- We can show that $\hat{\beta}_1 = \frac{Cov(x,y)}{Var(x)}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ minimizes the RSS.
- Bias of an estimator: if $\beta_i = E[\hat{\beta}_i]$, then the estimator is said to be unbiased. The least squares estimator is unbiased (under the assumption that $E[e_i] = 0$).
- To find out how far our predicted value is from the actual value of the estimator, we use the standard error (S.E) which is the square root of the variance of the estimator. Greater the SE, more is the noise associated with the measurement of the estimator.
- $SE(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$, $SE(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- For a gaussian distribution, there is a 95% chance that β_i lies in the interval $[\hat{\beta}_i - 2SE(\hat{\beta}_i), \hat{\beta}_i + 2SE(\hat{\beta}_i)]$. We get this by computing $\int_{\mu-2\sigma}^{\mu+2\sigma} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 0.9544$.
- If we wish to know if a linear relation between X and Y exists, we can test $H_0: \beta_1 = 0$ using the t-statistic. The t-statistic is given by $\hat{\beta}_i / SE(\hat{\beta}_i)$. The higher the t-statistic, greater the effect it has on the model.

- If an estimator significantly affects a model, it should have a high value and a low noise, corresponding to a higher t-statistic.
- If the p-value associated with the t-statistic is small, we reject the null hypothesis, we conclude that the predictor has an effect on the response.
- $RSS/(n-p-1)$ is an unbiased estimator of σ^2 . The square root of this quantity is called $RSE = \sqrt{\frac{RSS}{n-p-1}}$ where p is the number of predictors. It is used to assess the accuracy of the model. It is a measure of the lack of fit of the model to the data. Lower the RSE, better the accuracy of the model.
- $R^2 = 1 - RSS/TSS$ is also used to measure the accuracy of the model. It is a measure of how well the model represents the data compared to how well the mean represents the data. If the model represents the data much better than the mean, R^2 is close to 1. If the model does not fare any better than the mean, R^2 is close to 0.

Multiple Regression:

- More than one predictor
- It takes the form: $y = \beta_0 + \sum_{i=1}^p \beta_i X_i$ for p estimators.
- We can write it as $\hat{y} = X\hat{\beta}$. If we minimize the RSS ($|y - X\hat{\beta}|^2$), we get $\hat{\beta}$ as $(X^T X)^{-1} X^T y$
- The covariance matrix $\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$ has its i,jth element as $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j)$ and its diagonal elements as $\text{Var}(\hat{\beta}_i)$. This is under the assumption that $E[e_i] = 0$, $\text{Var}(e_i) = \sigma^2$ for all i and $\text{Cov}(e_i, e_j) = 0$. This can be used to calculate the SE, which can be used to calculate the t-statistic ($\hat{\beta}_i / SE(\hat{\beta}_i)$).
- As we add more predictors, the t-statistic may change if collinearity is introduced.
- If we wish to figure out if there exists a relation between the predictors and the response, we can use the F-test.
- To test if q predictors have an effect on the response, we test the null hypothesis $H_0: \hat{\beta}_i = 0$ for all q $\hat{\beta}_i$. We fit all the variables except these q variables. From this model, we calculate RSS_0 .
- F-statistic = $\frac{(RSS_0 - RSS)/q}{RSS/(n-p-1)}$. A high f-statistic implies that the q predictor variables have a significant effect on the response.
- The intuition behind this formula is that $(RSS_0 - RSS)/RSS$ is a measure of how good the model with all the variables is compared to the model without the q variables. Higher the value, better the original model. It is divided by a factor of q to avoid overfitting.

- If we wish to check if all predictor variables have an effect on the response, we use: $F\text{-statistic} = \frac{(TSS-RSS)/p}{RSS/(n-p-1)}$
- $E[RSS/(n-p-1)] = \sigma^2$. If the null hypothesis is true, $E[(TSS-RSS)/p] = \sigma^2$, and the f-statistic is around 1. If the null hypothesis is false, $E[(TSS-RSS)/p] > \sigma^2$, leading to a high f-statistic.
- Why do we need f-tests as compared to individual t-tests? If we have 100 predictor variables, each with a null hypothesis: $H_0 : \beta_i = 0$, and each null hypothesis is true, then there is a 0.05 probability that $p \leq 0.05$ for each p-value. The probability that at least one p value is lesser than or equal to 0.05 is given by $1 - (1 - 0.05)^{100} = 0.994$. So, the probability of at least one false positive is 0.994. If we use a single f-test, and the null hypothesis is true, the probability of a false positive is only 0.05. So, f-tests reduce the chance of a false-positive.
- If $p > n$, the least squares solution is not unique and leads to overfitting. We can use subset selection instead.
- For p estimators, there are 2^p possible models formed using subsets of the estimators. We choose a subset as some estimators may be redundant. These models can be compared using various statistics. Comparing all 2^p models is not usually feasible, so we use either forward selection, backward selection or mixed selection.
- In forward selection, we start with a null model. We see which of the p predictors improves the model the most (by comparing R^2). We add this to our model. We then check which of the remaining p-1 predictors improves our model the most and we add it to our model. Continuing this, we get p models. We compare these p models on various statistics and choose the best model.
- In backward selection, we start with a model with all variables and remove the variable with the highest p-value. This is done till a stopping rule is met. For example, all the remaining p-values should be lesser than a certain value.
- In mixed selection, we first implement forward selection. If any of the parameters has a p-value greater than some threshold, we implement backward selection on that parameter. We stop when no more variables can be added and all p-values are below a certain threshold.
- We can use RSE and R^2 to measure the model accuracy.
- In order to quantify the uncertainty of our model, and of our predictions, we use intervals.
- 95% confidence intervals have a 0.95 probability of having the value of $f(X)$ in the interval.

- 95% prediction intervals have a 0.95 probability of having the value of $\hat{Y}$ in the interval.